

KUNAL M SHENDE

Software Engineer | IoT & Embedded Systems Developer | Cybersecurity Researcher

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PROFESSIONAL SUMMARY

Results-driven Computer Science undergraduate with 3+ years of hands-on experience in embedded systems security, IoT hardware architecture, and full-stack software development. Architected ZeroTrace, an ESP32-S3-based cybersecurity audit companion featuring real-time 2.4 GHz RF spectrum analysis and native USB-HID automation, earning 2nd Prize among 50+ teams at TECHTANTRA 2K26 National Technical Competition. Secured an Indian Design Patent (Application No: 499678-001). Completed 5 industry internships spanning cybersecurity, cloud infrastructure, full-stack development, and mobile engineering. Proficient in 7+ programming languages and 15+ frameworks across hardware and software domains.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering

2023 - 2027

S. B. Jain Institute of Technology, Management and Research, Nagpur

Specialization: Software Development, IoT Architecture, Cybersecurity and Embedded Systems

Higher Secondary Certificate (HSC) in Science

2021 - 2023

Municipal Junior College, Kalmeshwar, Maharashtra State Board

Secondary School Certificate (SSC) with Distinction

2019 - 2021

Municipal Junior College, Kalmeshwar, Maharashtra State Board

TECHNICAL SKILLS

Programming Languages: C, Embedded C, C++, Python, JavaScript (ES6+), TypeScript, Bash

IoT & Hardware: ESP32, ESP32-S3, Arduino (ATmega328P), Raspberry Pi, FreeRTOS, Arduino Cloud, Blynk IoT, SPI, I2C, UART, ADC, Cisco Packet Tracer

Frontend: HTML5, CSS3, JavaScript, React.js, Next.js, Bootstrap, Tailwind CSS

Backend: Python Flask, Node.js, Express.js, WebSockets, Firebase, REST APIs

Databases: MySQL, PostgreSQL, Supabase, Firebase Firestore, SQLite

Cybersecurity: VAPT, Ethical Hacking, Network Auditing, Burp Suite, Nmap, OWASP Top 10, Wireshark, Metasploit

DevOps & Tools: Git, GitHub, Docker, Vercel, Render.com, Google Cloud Platform, VS Code, Figma, Postman, Notion

Languages: English (Professional), Hindi (Native), Marathi (Native), German (Level 1 Certified)

PROFESSIONAL EXPERIENCE

Independent Security Researcher, IoT & Hardware Security

2024 - Present

Freelance / Independent

- Built 4 custom IoT security tools on ESP32-S3 and ATmega328P platforms, reducing firmware vulnerability exposure by 40% through proactive system hardening.
- Conducted offensive security assessments across 10+ test environments, identifying and documenting 15+ critical vulnerabilities using OWASP and NIST methodologies.
- Performed network traffic audits using Wireshark and Nmap, decreasing security anomalies by 25% through targeted firewall rule optimization.
- Authored 3 technical write-ups on HID injection detection algorithms and RF spectrum analysis, contributing to open-source security research.

Web Exploit Hunting & Bug Bounty Intern

Jun 2026 - Aug 2026

EduSkills & AICTE

- Executed penetration tests on 8+ web applications, uncovering 12 security misconfigurations including SQL injection and broken access control flaws.
- Reported 5 high-severity vulnerabilities through responsible disclosure, applying Burp Suite and manual testing against OWASP Top 10 attack vectors.

Cybersecurity & SecOps Intern

Jan 2026 - Mar 2026

Palo Alto Networks / EduSkills

- Analyzed 200+ packet captures and firewall configurations, strengthening endpoint protection policies across 3 multi-cloud environments.
- Completed 4 Palo Alto Networks certifications (SecOps, Network Security, Cybersecurity, Cloud Security) within 12 weeks.

Java Full Stack Developer Intern

Oct 2025 - Dec 2025

EduSkills & AICTE

- Constructed 3 full-stack web applications with Java backend services, RESTful APIs, and relational database integration, serving 500+ concurrent users.
- Streamlined authentication workflows, cutting login response time by 30% through session caching and middleware optimization.

Cloud Architect Intern

Apr 2025 - Jun 2025

AWS Academy

- Provisioned cloud infrastructure using EC2, S3, VPC, and IAM, supporting 99.9% uptime for 2 production-grade web deployments.
- Automated resource provisioning scripts, reducing manual setup time by 50% across staging and production environments.

Android Developer Intern

Jan 2025 - Mar 2025

Google for Developers

- Launched 2 native Android applications with Material Design UI, background sensor services, and offline-first SQLite caching for 1,000+ potential users.
- Integrated RESTful client communication layers, achieving sub-200ms API response times through asynchronous task management.

KEY PROJECTS

ZeroTrace: CyberSecurity Audit & Multipurpose Device

2026

ESP32-S3 | FreeRTOS | C++ | ST7735 TFT | USB-HID | github.com/KunalShende1436/ZeroTrace---CyberSecurity-Audit-Multipurpose-Device

- Architected a dual-core embedded companion on ESP32-S3 with ST7735 128x160 color TFT display, achieving 60 FPS non-blocking UI rendering through optimized SPI bus scheduling.
- Integrated 4 security modules: 2.4 GHz RF spectrum scanner, Wi-Fi network auditor, RSSI foxhunter, and native USB-HID keystroke automation engine.
- Won 2nd Prize among 50+ competing teams at TECHTANTRA 2K26 National Technical Competition; secured Indian Design Patent (Application No: 499678-001, CBR No: 209757).

Sentinel: HID Attack Detection Framework

2026

Python | Windows API | WMI | Tkinter | Real-Time Monitoring

- Created a modular detection system that identifies rogue USB devices and keystroke injection anomalies with sub-10ms mitigation latency.
- Incorporated forensic snapshotting and a severity-based GUI dashboard, processing 100+ HID events per second for real-time incident alerting.

MicroFlasher: Protected Binary Web Serial Flashing Platform

2026

TypeScript | Next.js | Web Serial API | WebSockets | ESP32

- Produced a zero-install browser-based firmware flashing utility, eliminating native toolchain setup for 3 microcontroller families (ESP32, ESP8266, ATmega).
- Applied chunked binary transfer protocols with AES encryption, maintaining 98% upload success rate across 50+ flashing sessions.

Dual-Band RF Denial & Signal Isolation Appliance

2026

RF Hardware | 2.4 GHz / Bluetooth | Signal Analysis

- Constructed an experimental RF isolation system for wireless resilience testing across 2.4 GHz Wi-Fi and Bluetooth spectrums, validating failover behavior in 5 IoT mesh networks.
- Autonomous Line-Follower Robot** 2025
- ATmega328P | L298N | IR Sensors | TCCR0B PWM | [github.com/KunalShende1436/Autonomous-Line-Follower-Robot](#)*
- Fabricated a differential drive robotics platform with hardware-level PWM calibration (TCCR0B at 7.8 kHz), eliminating motor stalling at low-duty cycles.
 - Tuned a closed-loop proportional controller with 5-sensor IR array, achieving 95% accuracy on tracks with sharp 90-degree corners.

- Chai & Kafe: Artisan Indian Bistro Web Platform** 2025
- HTML5 | CSS3 | JavaScript | Bootstrap | Responsive Design*
- Delivered a responsive restaurant web platform with fluid menu interactions, reservation workflows, and mobile-first layout serving 300+ monthly visitors.

ACHIEVEMENTS & AWARDS

- Earned 2nd Prize at TECHTANTRA 2K26 National Level Technical Project Competition among 50+ teams for the ZeroTrace cybersecurity hardware project.
- Secured Indian Design Patent (Application No: 499678-001, CBR No: 209757) for ZeroTrace hardware enclosure, filed April 2026.

CERTIFICATIONS

Cyber Security Workshop - <i>Crow Security and IIT Bombay</i>	<i>2025 - 2026</i>
NASSCOM Certified Ethical Hacking - <i>NASSCOM FutureSkills Prime</i>	<i>2025 - 2026</i>
Security Operations Fundamentals - <i>Palo Alto Networks</i>	<i>March 2026</i>
Network Security Fundamentals - <i>Palo Alto Networks</i>	<i>March 2026</i>
Cybersecurity Fundamentals - <i>Palo Alto Networks</i>	<i>March 2026</i>
Cloud Security Fundamentals - <i>Palo Alto Networks</i>	<i>March 2026</i>
Web Exploit Hunting Virtual Internship - <i>AICTE and EduSkills</i>	<i>Jun - Aug 2026</i>
Cybersecurity Virtual Internship - <i>EduSkills and AICTE</i>	<i>Jan - Mar 2026</i>
Java Full Stack Developer Internship - <i>EduSkills and AICTE</i>	<i>Oct - Dec 2025</i>
AWS Cloud Computing Fundamentals - <i>AWS Academy</i>	<i>Apr - Jun 2025</i>
Android Developer Internship - <i>Google for Developers</i>	<i>Jan - Mar 2025</i>
Python Basic and Problem Solving - <i>HackerRank</i>	<i>2025</i>
German Language Proficiency Level 1 - <i>Foreign Language Examination Board</i>	<i>December 2025</i>